

COMP 7370/7370-D01 — Advanced Computer and Network Security

Homework Assignment Solutions

Note: a=0, b=1, c=2, ... z=25 throughout.

Question 1 — Affine Cipher Encryption: $c = 5m + 7 \pmod{26}$, plaintext = "helpme"

Apply $c = 5m + 7 \pmod{26}$ letter by letter:

Letter	m	$5m + 7$	$5m + 7 \pmod{26}$	Ciphertext
h	7	42	16	q
e	4	27	1	b
l	11	62	10	k
p	15	82	4	e
m	12	67	15	p
e	4	27	1	b
				qbkepb

Answer: Ciphertext = "qbkepb"

Question 2 — Affine Cipher Decryption: $c = 11m + 2 \pmod{26}$, ciphertext = "vmwz"

Decryption: $m = 11^{-1}(c - 2) \pmod{26}$. First find $11^{-1} \pmod{26}$:

Extended Euclidean: $\gcd(11, 26) \rightarrow 11 \times 19 = 209 = 8 \times 26 + 1 \equiv 1 \pmod{26} \rightarrow 11^{-1} = 19$

Letter	c	$c - 2$	$19 \times (c - 2)$	$19 \times (c - 2) \pmod{26}$	Plaintext
v	21	19	361	23	x
m	12	10	190	8	i
w	22	20	380	16	q
z	25	23	437	21	v
					xiqv

Answer: Plaintext = "xiqv"

Question 3 — Hill Cipher Key Recovery

Given: plaintext = "friday", ciphertext = "pqcfku", block size 2. Encryption: $C = MK \pmod{26}$.

Use the first two blocks: fr $\rightarrow$ pq and id $\rightarrow$ cf.

$$M = \begin{bmatrix} 5 & 17 \\ 8 & 3 \end{bmatrix} \quad C = \begin{bmatrix} 15 & 16 \\ 2 & 5 \end{bmatrix}$$

$$K = M^{-1} \cdot C \pmod{26}$$

$$\det(M) = 5 \times 3 - 17 \times 8 = 15 - 136 = -121 \equiv -121 + 5 \times 26 = 9 \pmod{26}$$

$$9^{-1} \pmod{26}: \text{extended Euclidean} \rightarrow 9 \times 3 = 27 \equiv 1 \pmod{26} \rightarrow \det^{-1} = 3$$

$$M^{-1} = 3 \times [[3, -17], [-8, 5]] \bmod 26 = [[9, 1], [2, 15]] \bmod 26$$

$$K = M^{-1} \times C = [[9, 1], [2, 15]] \times [[15, 16], [2, 5]] \bmod 26$$

$$K[0][0] = (9 \times 15 + 1 \times 2) \bmod 26 = 137 \bmod 26 = 7$$

$$K[0][1] = (9 \times 16 + 1 \times 5) \bmod 26 = 149 \bmod 26 = 19$$

$$K[1][0] = (2 \times 15 + 15 \times 2) \bmod 26 = 60 \bmod 26 = 8$$

$$K[1][1] = (2 \times 16 + 15 \times 5) \bmod 26 = 107 \bmod 26 = 3$$

Verification with 3rd block 'ay' $\rightarrow$: $[0, 24] \times [[7, 19], [8, 3]] = [0 + 192, 0 + 72] = [192, 72] \equiv [10, 20] \bmod 26 \rightarrow$ 'ku' ✓

Answer: K = [[7, 19], [8, 3]] (i.e., 'h', 't', 'i', 'd')

Question 4 — Extended Euclidean Algorithm (Multiplicative Inverses)

(a) $17^{-1} \bmod 101$

q	r	s	t
	17	1	0
	101	0	1
0	101	0	1
5	17	1	0
1	16	-5	1
16	1	6	-1

$17^{-1} \bmod 101 = 6$ (check: $17 \times 6 \bmod 101 = 1$)

(b) $357^{-1} \bmod 1234$

q	r	s	t
	357	1	0
	1234	0	1
0	1234	0	1
3	357	1	0
2	163	-3	1
5	31	7	-2
3	8	-38	11
1	7	121	-35
7	1	-159	46

$357^{-1} \bmod 1234 = 1075$ (check: $357 \times 1075 \bmod 1234 = 1$)

(c) $3125^{-1} \bmod 9987$

q	r	s	t
	3125	1	0
	9987	0	1

0	9987	0	1
3	3125	1	0
5	612	-3	1
9	65	16	-5
2	27	-147	46
2	11	310	-97
2	5	-767	240
5	1	1844	-577

$3125^{-1} \bmod 9987 = 1844$ (check: $3125 \times 1844 \bmod 9987 = 1$)

Question 5 — Shift Cipher Exhaustive Key Search

Ciphertext: BEEAKFYDJXUQYHYJIQRYHTYJIQFBQDUYJIIKFUHCQD

Program (Python):

```
ciphertext = "BEEAKFYDJXUQYHYJIQRYHTYJIQFBQDUYJIIKFUHCQD"
for k in range(26):
    decrypted = ''.join(chr((ord(c)-ord('A')-k)%26+ord('a'))) for c in ciphertext)
    print(f"k={k:2d}: {decrypted}")
```

All 26 trials (correct answer **highlighted**):

Key (k)	Decrypted Text
0	beeakfydjxuqyhyjiqryhtyjiqfbqduyjiikfuhcq
1	addzjexcwtpxgxihpqxsixhpeapctxihhjetgbpc
2	zccyidwbhvsowfwhgopwfrwhgodzobswhggidsfaob
3	ybbxhcavagurnvevgfnoveqvgfncynarvgffhcrezna
4	xaawgbuzftqmudufemnudpufembxmzqufeegbqmdymz
5	wzzvfatyesphtctedlmtcotedlawlypteddfapcxly
6	vyyuezsxdroksbsdcklsbnsdckzvksosdccezbwkw
7	uxxtdyrwcqnjrarcbjkrarcbjyujwnrcbbdynavjw
8	twwscxqvbpmiqzqbaijzqlbaixtivmqbaacxmzuiv
9	svvrbwpuaoahpypazhipykpazhwshulpazzbwlythu
10	ruuqavotznkgoxozyghoxjozygvrgtkozyavkxsgt
11	qttpzunsymjfnwnyxfgnwinyxfuqfsjnyxxzujwrfs
12	pssoytmrxliemvmxwefmvhmxwetperimxwwytivqer
13	orrxslqwkhdulwvdeluglwvdsodqhlwvwxshupdq
14	nqqmwrkpvjgcktkvucdktkvucrcnpgkvuuwrgtocp
15	mpplvjouifbjsjutbcjsejutmbofjuttvqfsnbo
16	lookupintheairitsabirditsaplaneitssuperman
17	knnjtohmgsdzhqhsrzahqchsrzokzmdhsrrtodqlzm
18	jmmisnglrfcygpgqyqzgbgrqynjylcgrqqsnpcpkl

19	illhrmfkqebxfqpxyfoafqpxmixkbfqpprmbojxk
20	hkkggqlejpdawenepowxenzepowlhwaepooqlaniwj
21	gjffpkdioczdmdonvwdmydonvkgvizdonnpkzmvhi
22	fiieojchnbyuclcnmuvclxcnmujfuhycnmnojylguh
23	ehhdnibgmaxtbkbmltubkwbmlietgxbmlinixkftg
24	dggcmhafizwsajalkstajvalkshdsfwalkkmhwjesf
25	cffblgzekyvrzizkjsziuzkjrgcrevzkjlgvidre

Answer: k = 16 → "lookupintheairitsabirditsaplaneitssuperman"

= "Look up in the air, it's a bird, it's a plane, it's Superman!"

Question 6 — Entropy Calculations

Setup: $P=\{a,b,c\}$, $K=\{K1,K2,K3\}$ (equiprobable), $C=\{1,2,3,4\}$.

$\Pr[a]=1/2$, $\Pr[b]=1/3$, $\Pr[c]=1/6$. Encryption matrix:

	a	b	c
K1	1	2	3
K2	2	3	4
K3	3	4	1

$$H(P) = -(1/2)\log_2(1/2) - (1/3)\log_2(1/3) - (1/6)\log_2(1/6)$$

$$= 0.5 + 0.5283 + 0.4308 = \mathbf{1.4591 \text{ bits}}$$

$$H(K) = \log_2(3) \text{ (equiprobable over 3 keys)}$$

$$= \mathbf{1.5850 \text{ bits}}$$

Ciphertext probabilities:

$$\Pr[C=1] = \Pr[K1,c] + \Pr[K3,a] = 1/3 \times 1/6 + 1/3 \times 1/2 = 1/18 + 3/18 = 2/9$$

$$\Pr[C=2] = \Pr[K1,b] + \Pr[K2,a] = 1/9 + 1/6 = 5/18$$

$$\Pr[C=3] = \Pr[K1,a] + \Pr[K2,b] + \Pr[K3,c] = 1/6 + 1/9 + 1/18 = 6/18 = 1/3$$

$$\Pr[C=4] = \Pr[K2,c] + \Pr[K3,b] = 1/18 + 1/9 = 3/18 = 1/6$$

$$H(C) = -(2/9)\log(2/9) - (5/18)\log(5/18) - (1/3)\log(1/3) - (1/6)\log(1/6)$$

$$= \mathbf{1.9547 \text{ bits}}$$

$$H(K|C) = \sum \Pr[C=c] \times H(K|C=c) = \mathbf{1.0894 \text{ bits}}$$

$$H(P|C) = \sum \Pr[C=c] \times H(P|C=c) = \mathbf{1.0894 \text{ bits}}$$

(Note: $H(P|C) = H(K|C)$ here because each key maps each plaintext to a unique ciphertext — the cipher is deterministic and one-to-one per key.)

Question 7 — Chinese Remainder Theorem

System: $x \equiv 12 \pmod{25}$, $x \equiv 9 \pmod{26}$, $x \equiv 23 \pmod{27}$

$$M = 25 \times 26 \times 27 = 17550$$

$$M_{25} = M/25 = 702, M_{26} = M/26 = 675, M_{27} = M/27 = 650$$

$$y_{25} = M_{25}^{-1} \bmod 25 = 702^{-1} \bmod 25 = 13 \text{ (check: } 702 \times 13 \bmod 25 = 1)$$

$$y_{26} = M_{26}^{-1} \bmod 26 = 675^{-1} \bmod 26 = 25 \text{ (check: } 675 \times 25 \bmod 26 = 1)$$

$$y_{27} = M_{27}^{-1} \bmod 27 = 650^{-1} \bmod 27 = 14 \text{ (check: } 650 \times 14 \bmod 27 = 1)$$

$$x = (12 \times 702 \times 13 + 9 \times 675 \times 25 + 23 \times 650 \times 14) \bmod 17550$$

$$= (109512 + 151875 + 209300) \bmod 17550$$

$$= 470687 \bmod 17550$$

Answer: $x = 14387$ (check: $14387 \bmod 25 = 12$ ✓, $\bmod 26 = 9$ ✓, $\bmod 27 = 23$ ✓)

Question 8 — Discrete Logarithm ($p=227$, $\alpha=2$, primitive root)

(a) Compute $2^{32}, 2^{40}, 2^{59}, 2^{156} \bmod 227$ and factor over $\{2,3,5,7,11\}$:

Exponent e	$2^e \bmod 227$	Factorization
2^{32}	176	$2^4 \times 11$
2^{40}	110	$2 \times 5 \times 11$
2^{59}	60	$2^2 \times 3 \times 5$
2^{156}	28	$2^2 \times 7$

(b) Compute discrete logs $\bmod (p-1)=226$, using $\log 2=1$:

$$\text{From } 2^{32} \equiv 2^4 \times 11: 32 = 4 \cdot \log 2 + \log 11 \rightarrow \log 11 = 32 - 4 = \mathbf{28}$$

$$\text{From } 2^{40} \equiv 2 \times 5 \times 11: 40 = \log 2 + \log 5 + \log 11 \rightarrow \log 5 = 40 - 1 - 28 = \mathbf{11}$$

$$\text{From } 2^{59} \equiv 2^2 \times 3 \times 5: 59 = 2 \cdot \log 2 + \log 3 + \log 5 \rightarrow \log 3 = 59 - 2 - 11 = \mathbf{46}$$

$$\text{From } 2^{156} \equiv 2^2 \times 7: 156 = 2 \cdot \log 2 + \log 7 \rightarrow \log 7 = 156 - 2 = \mathbf{154}$$

Verification:

$$2^{46} \bmod 227 = 3 \text{ (should be 3: } \checkmark)$$

$$2^{11} \bmod 227 = 5 \text{ (should be 5 mod factors... } \log 5=11: 2^{11} \bmod 227 = 5 \checkmark)$$

$$2^{154} \bmod 227 = 7 \text{ (should be 7: } \checkmark)$$

$$2^{28} \bmod 227 = 11 \text{ (should be 11: } \checkmark)$$

Question 9 — GCD(57, 93) via Extended Euclidean

Find $\gcd(57, 93)$ and integers s, t such that $57s + 93t = \gcd(57, 93)$.

q	r	s	t
	57	1	0
	93	0	1
0	93	0	1
1	57	1	0
1	36	-1	1
1	21	2	-1

1	15	-3	2
2	6	5	-3
2	3	-13	8

$\gcd(57, 93) = 3$

$s = -13, t = 8$

Verification: $57 \times (-13) + 93 \times (8) = -741 + 744 = 3 \checkmark$